

A Note on OptNet

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Abstract

This paper provides a detailed derivation of the results in Section 3 of *OptNet: Differentiable Optimization as a Layer in Neural Networks* [1].

1 Introduction

There has been a strong interest in embedding optimization problems in the layers of neural networks, enabling the learning of the structure of the problems through end-to-end training. Consider the following constrained quadratic optimization problem:

$$\begin{aligned} & \underset{z}{\text{minimize}} && \frac{1}{2}z^T Q z + q^T z \\ & \text{subject to} && Az = b, \\ & && Gz \leq h, \end{aligned} \tag{1}$$

where $z \in \mathbb{R}^n$ is optimization variables, $Q \in \mathbb{R}^{n \times n}$ ($Q \succeq 0$), $q \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $G \in \mathbb{R}^{p \times n}$ and $h \in \mathbb{R}^p$ are parameters of the optimization problems. This optimization problem can be treated as a layer in neural network, where the optimization objective and constraints can be expressed in a way that cannot be accomplished by fully connected layers. To achieve end-to-end training of neural networks with such layer, it is necessary to conduct both forward and backward passes. While forward passes can be achieved by solving the optimization problem using a quadratic programming solver, executing backward passes is not as straightforward. Modern neural network training relies on employing backpropagation and autodifferentiation, which necessitates the storing of intermediate results obtained during forward pass for the gradient calculations. When solving optimization problem becomes a part of forward pass, the iterative nature of various optimization problem solvers will incur numerous intermediate results, such as Jacobian matrix or Hessian matrix with respect to the optimization variables. The storage and manipulation of these intermediate results lead to excessive memory usage and numerical instability, making backward pass and end-to-end training intractable. Amos and Kolter [1] provide a novel solution to this problem, by leveraging implicit function theorem, they demonstrated that the gradients of loss function with respect to the optimization parameters can be obtained regardless of the solver used in the forward pass. This note provides a detailed explanation of how implicit function theorem and optimization theory can be used to obtain these gradients. Knowledge of implicit function theorem

and optimization theory are assumed for readers. Readers who are unfamiliar with the implicit function theorem may refer to Chapter 2 of the tutorial by Kolter et al. [2].

2 The Implicit Function

We started by noticing problem (1) is a convex optimization problem and the KKT conditions are sufficient for global optimality. Thus, at optimality, we have

$$\begin{bmatrix} Qz^* + q + A^T v^* + G^T \lambda^* \\ D(\lambda^*)(Gz^* - h) \\ Az^* - b \end{bmatrix} = \mathbf{0}, \quad (2)$$

where $D(\cdot)$ creates a diagonal matrix from a vector and z^*, λ^*, v^* , are the optimal primal and dual variables. Observe that $\{Q, q, A, b, G, h\}$ are parameters of the above vector-valued function, and for each different set of parameters, there exists a unique set of optimal primal and dual variables. By leveraging the implicit function theorem, an implicit function $F(x, \theta^*(x))$ can be defined as

$$F(x, \theta^*(x)) = \begin{bmatrix} Qz^* + q + A^T v^* + G^T \lambda^* \\ D(\lambda^*)(Gz^* - h) \\ Az^* - b \end{bmatrix} \in \mathbb{R}^{n+m+p}, \quad (3)$$

with x and $\theta^*(x)$ defined as

$$\begin{aligned} x &= [\text{vec}(Q)^T, q^T, \text{vec}(A)^T, b^T, \text{vec}(G)^T, h^T]^T \in \mathbb{R}^{n^2+n+mn+m+pn+p} \\ \theta^*(x) &= [z^*(x), \lambda^*(x), v^*(x)]^T \in \mathbb{R}^{n+m+p}. \end{aligned} \quad (4)$$

To align with the function definition in the implicit function theorem and to avoid dealing with tensors in matrix differentiation, we have vectorized the matrices to vectors in a column-major order. For example, matrix Q is expressed as a column vector $\text{vec}(Q) = [Q_{11}, Q_{21} \dots Q_{n1}, Q_{12}, \dots Q_{n2} \dots Q_{1n} \dots Q_{nn}]^T \in \mathbb{R}^{n^2}$. Given that $F(x, \theta^*(x)) = \mathbf{0}$ at optimality, by differentiating both sides with respect to x we obtain

$$\frac{\partial F(x, \theta^*(x))}{\partial x} = \frac{\partial F(x, \theta^*)}{\partial x} + \frac{\partial F(x, \theta^*)}{\partial \theta^*} \frac{\partial \theta^*(x)}{\partial x} = \mathbf{0}. \quad (5)$$

The essential components $\frac{\partial \theta^*(x)}{\partial x}$ required for gradient computation can therefore be obtained by

$$\frac{\partial \theta^*(x)}{\partial x} = - \left(\frac{\partial F(x, \theta^*)}{\partial \theta^*} \right)^{-1} \frac{\partial F(x, \theta^*)}{\partial x}. \quad (6)$$

3 Component Derivation

In this section, we provide detailed derivation for each component of equation (6). By applying basic matrix calculus results, we immediately obtain

$$\frac{\partial F(x, \theta^*)}{\partial \theta^*} = \begin{bmatrix} Q & G^T & A^T \\ D(\lambda^*)G & D(Gz^* - h) & 0 \\ A & 0 & 0 \end{bmatrix}. \quad (7)$$

The exact form of $\frac{\partial \theta^*(x)}{\partial x}$ cannot be directly derived due to its implicit nature, and the derivation of $\frac{\partial \theta^*(x)}{\partial x}$ requires a component-wise treatment. We find it beneficial to expand them in vector and matrix formats to aid readers' understanding of subsequent content. Therefore, we express $\frac{\partial \theta^*(x)}{\partial x}$ as

$$\frac{\partial \theta^*(x)}{\partial x} = \begin{bmatrix} \frac{\partial z^*(x)}{\partial x} \\ \frac{\partial \lambda^*(x)}{\partial x} \\ \frac{\partial v^*(x)}{\partial x} \end{bmatrix} = \begin{bmatrix} \frac{\partial z^*(x)}{\partial \text{vec}(Q)} & \dots & \frac{\partial z^*(x)}{\partial h} \\ \frac{\partial \lambda^*(x)}{\partial \text{vec}(Q)} & \dots & \frac{\partial \lambda^*(x)}{\partial h} \\ \frac{\partial v^*(x)}{\partial \text{vec}(Q)} & \dots & \frac{\partial v^*(x)}{\partial h} \end{bmatrix}. \quad (8)$$

Similarly, the vector and matrix expansion for $\frac{\partial F(x, \theta^*)}{\partial x}$ is expressed as

$$\frac{\partial F(x, \theta^*)}{\partial x} = \begin{bmatrix} \frac{\partial F_1(x, \theta^*)}{\partial x} \\ \frac{\partial F_2(x, \theta^*)}{\partial x} \\ \frac{\partial F_3(x, \theta^*)}{\partial x} \end{bmatrix} = \begin{bmatrix} \frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(Q)} & \dots & \frac{\partial F_1(x, \theta^*)}{\partial h} \\ \frac{\partial F_2(x, \theta^*)}{\partial \text{vec}(Q)} & \dots & \frac{\partial F_2(x, \theta^*)}{\partial h} \\ \frac{\partial F_3(x, \theta^*)}{\partial \text{vec}(Q)} & \dots & \frac{\partial F_3(x, \theta^*)}{\partial h} \end{bmatrix}, \quad (9)$$

where $F_1(x, \theta^*)$, $F_2(x, \theta^*)$, and $F_3(x, \theta^*)$ correspond to the 1_{st}, 2_{nd}, and 3_{rd} rows of function $F(x, \theta^*(x))$ in (3). Now, we perform derivation for each component of matrix (9).

3.1 $F_1(x, \theta^*)$

We immediately notice that in the 1_{st} row of matrix $\frac{\partial F(x, \theta^*)}{\partial x}$

$$\begin{aligned} \frac{\partial F_1(x, \theta^*)}{\partial q} &= \frac{\partial q}{\partial q} = I \\ \frac{\partial F_1(x, \theta^*)}{\partial b} &= \frac{\partial F_1(x, \theta^*)}{\partial h} = 0. \end{aligned} \quad (10)$$

$\frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(Q)}$ can be expressed as

$$\frac{\partial Q z^*}{\partial \text{vec}(Q)} = \begin{bmatrix} \frac{\partial (Q z^*)_1}{\partial \text{vec}(Q)} \\ \vdots \\ \frac{\partial (Q z^*)_n}{\partial \text{vec}(Q)} \end{bmatrix} = \begin{bmatrix} \frac{\partial (Q z^*)_1}{\partial Q_{11}} & \dots & \frac{\partial (Q z^*)_1}{\partial Q_{nn}} \\ \vdots & \ddots & \vdots \\ \frac{\partial (Q z^*)_n}{\partial Q_{11}} & \dots & \frac{\partial (Q z^*)_n}{\partial Q_{nn}} \end{bmatrix}. \quad (11)$$

The partial derivative of each component of this matrix can be expressed compactly as

$$\frac{\partial (Q z^*)_i}{\partial Q_{jk}} = \frac{\partial \sum_{\ell=1}^n Q_{i\ell} z_\ell^*}{\partial Q_{jk}} = \begin{cases} z_k^* & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}, \quad (12)$$

resulting in a matrix of form

$$\begin{bmatrix} z_1^* & 0 & \dots & 0 & \dots & z_n^* & 0 & \dots & 0 \\ 0 & z_1^* & \dots & 0 & \dots & 0 & z_n^* & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & z_1^* & \dots & 0 & 0 & \dots & z_n^* \end{bmatrix}. \quad (13)$$

By using Kronecker product, we can express this matrix as

$$\frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(Q)} = (z^*)^T \otimes I_n \in \mathbb{R}^{n \times n^2}. \quad (14)$$

$\frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(A)}$ can be expressed as

$$\frac{\partial A^T v^*}{\partial \text{vec}(A)} = \begin{bmatrix} \frac{\partial(A^T v^*)_1}{\partial \text{vec}(A)} \\ \vdots \\ \frac{\partial(A^T v^*)_n}{\partial \text{vec}(A)} \end{bmatrix} = \begin{bmatrix} \frac{\partial(A^T v^*)_1}{\partial A_{11}} & \dots & \frac{\partial(A^T v^*)_1}{\partial A_{mn}} \\ \vdots & \ddots & \vdots \\ \frac{\partial(A^T v^*)_n}{\partial A_{11}} & \dots & \frac{\partial(A^T v^*)_n}{\partial A_{mn}} \end{bmatrix}. \quad (15)$$

The partial derivative of each component of this matrix can be expressed compactly as

$$\frac{\partial(A^T v^*)_i}{\partial A_{jk}} = \frac{\partial \sum_{\ell=1}^m A_{i\ell}^T v_\ell^*}{\partial A_{jk}} = \frac{\partial \sum_{\ell=1}^m A_{\ell i} v_\ell^*}{\partial A_{jk}} = \begin{cases} v_j^* & \text{if } i = k \\ 0 & \text{otherwise} \end{cases}, \quad (16)$$

resulting in a matrix of form

$$\begin{bmatrix} v_1^* & \dots & v_m^* & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ 0 & \dots & 0 & v_1^* & \dots & v_m^* & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 0 & \dots & 0 & \dots & v_1^* & \dots & v_m^* \end{bmatrix}. \quad (17)$$

By using Kronecker product, we can express this matrix as

$$\frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(A)} = I_n \otimes (v^*)^T \in \mathbb{R}^{n \times mn}. \quad (18)$$

Using the same derivation procedure of $\frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(A)}$, we can show that

$$\frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(G)} = I_n \otimes (\lambda^*)^T \in \mathbb{R}^{n \times pn}. \quad (19)$$

3.2 $F_2(x, \theta^*)$

It can be readily seen that

$$\begin{aligned} \frac{\partial F_2(x, \theta^*)}{\partial \text{vec}(Q)} &= \frac{\partial F_2(x, \theta^*)}{\partial q} = \frac{\partial F_2(x, \theta^*)}{\partial \text{vec}(A)} = \frac{\partial F_2(x, \theta^*)}{\partial b} = 0 \\ \frac{\partial F_2(x, \theta^*)}{\partial h} &= \frac{\partial D(\lambda^*)h}{\partial h} = -D(\lambda^*). \end{aligned} \quad (20)$$

$\frac{\partial F_2(x, \theta^*)}{\partial \text{vec}(G)}$ can be expressed as

$$\frac{\partial D(\lambda^*)Gz^*}{\partial \text{vec}(G)} = \begin{bmatrix} \frac{\partial \lambda_1^*(Gz^*)_1}{\partial \text{vec}(G)} \\ \vdots \\ \frac{\partial \lambda_p^*(Gz^*)_p}{\partial \text{vec}(G)} \end{bmatrix} = \begin{bmatrix} \frac{\partial \lambda_1^*(Gz^*)_1}{\partial G_{11}} & \dots & \frac{\partial \lambda_1^*(Gz^*)_1}{\partial G_{pn}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \lambda_p^*(Gz^*)_p}{\partial G_{11}} & \dots & \frac{\partial \lambda_p^*(Gz^*)_p}{\partial G_{pn}} \end{bmatrix}. \quad (21)$$

The partial derivative of each component of this matrix can be expressed compactly as

$$\frac{\partial \lambda_i^*(Gz^*)_i}{\partial G_{jk}} = \frac{\partial \lambda_i^* \sum_{\ell=1}^n G_{i\ell} z_\ell^*}{\partial G_{jk}} = \begin{cases} \lambda_j^* z_k^* & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}, \quad (22)$$

resulting in a matrix of form

$$\begin{bmatrix} \lambda_1^* z_1^* & 0 & \cdots & 0 & \cdots & \lambda_1^* z_n^* & 0 & \cdots & 0 \\ 0 & \lambda_2^* z_1^* & \cdots & 0 & \cdots & 0 & \lambda_2^* z_n^* & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_p^* z_1^* & \cdots & 0 & 0 & \cdots & \lambda_p^* z_n^* \end{bmatrix}. \quad (23)$$

By using Kronecker product, we can express this matrix as

$$\frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(G)} = (z^*)^T \otimes D(\lambda^*) \in \mathbb{R}^{p \times pn}. \quad (24)$$

3.3 $F_3(x, \theta^*)$

It can be readily seen that

$$\begin{aligned} \frac{\partial F_3(x, \theta^*)}{\partial \text{vec}(Q)} &= \frac{\partial F_3(x, \theta^*)}{\partial q} = \frac{\partial F_3(x, \theta^*)}{\partial \text{vec}(G)} = \frac{\partial F_3(x, \theta^*)}{\partial h} = 0 \\ \frac{\partial F_3(x, \theta^*)}{\partial b} &= -\frac{\partial b}{\partial b} = -I. \end{aligned} \quad (25)$$

By employing the same derivation procedure of $\frac{\partial F_1(x, \theta^*)}{\partial \text{vec}(Q)}$, we can also show that

$$\frac{\partial F_3(x, \theta^*)}{\partial \text{vec}(A)} = (z^*)^T \otimes I_m \in \mathbb{R}^{m \times mn}. \quad (26)$$

3.4 Summary

By incorporating everything derived from Section 3.1 to 3.3, we obtain following $(n + p + m) \times (n^2 + n + mn + m + pn + p)$ matrix

$$\frac{\partial F(x, \theta^*)}{\partial x} = \begin{bmatrix} (z^*)^T \otimes I_n & I_n & I_n \otimes (v^*)^T & 0 & I_n \otimes (\lambda^*)^T & 0 \\ 0 & 0 & 0 & 0 & (z^*)^T \otimes D(\lambda^*) & -D(\lambda^*) \\ 0 & 0 & (z^*)^T \otimes I_m & -I_m & 0 & 0 \end{bmatrix}, \quad (27)$$

where, from left to right, each column corresponds to $\frac{\partial F(x, \theta^*)}{\partial \text{vec}(Q)}$, $\frac{\partial F(x, \theta^*)}{\partial q}$, $\frac{\partial F(x, \theta^*)}{\partial \text{vec}(A)}$, $\frac{\partial F(x, \theta^*)}{\partial b}$, $\frac{\partial F(x, \theta^*)}{\partial \text{vec}(G)}$, and $\frac{\partial F(x, \theta^*)}{\partial h}$, respectively.

4 Gradient Computation

The gradients of loss function ℓ with respect to optimization problem parameters x are computed by employing chain rule. Assuming all vectors are column vectors, the gradient can be expressed as

$$\begin{aligned} \nabla_x \ell &= \left(\frac{\partial \ell}{\partial \theta^*} \right)^T \frac{\partial \theta^*(x)}{\partial x} \quad (\text{row vector}) \\ &= \left(\frac{\partial \theta^*(x)}{\partial x} \right)^T \frac{\partial \ell}{\partial \theta^*} \quad (\text{column vector}) \\ &= - \left(\frac{\partial F(x, \theta^*)}{\partial x} \right)^T \left(\frac{\partial F(x, \theta^*)}{\partial \theta^*} \right)^{-T} \frac{\partial \ell}{\partial \theta^*}, \end{aligned} \quad (28)$$

where the last equality is obtained by employing result obtained in equation (6) and $\frac{\partial \ell}{\partial \theta^*} = [\frac{\partial \ell}{\partial z^*}, \frac{\partial \ell}{\partial \lambda^*}, \frac{\partial \ell}{\partial v^*}]^T \in \mathbb{R}^{n+p+m}$.

During forward pass, only the output of optimization problem layer, i.e., the optimal primal variables z^* , is fed to next layer, therefore there are only gradients with respect to z^* and $\frac{\partial \ell}{\partial \theta^*} = [\frac{\partial \ell}{\partial z^*}, 0, 0]$. We define an intermediate vector $[d_z, d_\lambda, d_v]^T$ to represent the product of last two terms of equation (28), i.e., $(\frac{\partial F(x, \theta^*)}{\partial \theta^*})^{-T} \frac{\partial \ell}{\partial \theta^*}$. Using the results obtained in Section 3, the intermediate vector can be computed as

$$\begin{bmatrix} d_z \\ d_\lambda \\ d_v \end{bmatrix} = - \begin{bmatrix} Q & G^T & D(\lambda^*) \\ G & D(Gz^* - h) & 0 \\ A & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\partial \ell}{\partial z^*} \\ 0 \\ 0 \end{bmatrix}, \quad (29)$$

which corresponds to equation (7) of the paper. In addition, the transpose of $\frac{\partial F(x, \theta^*)}{\partial x}$ can be expressed as

$$(\frac{\partial F(x, \theta^*)}{\partial x})^T = \begin{bmatrix} z^* \otimes I_n & 0 & 0 \\ I_n & 0 & 0 \\ I_n \otimes v^* & 0 & z^* \otimes I_m \\ 0 & 0 & -I_m \\ I_n \otimes \lambda^* & z^* \otimes D(\lambda^*) & 0 \\ 0 & -D(\lambda^*) & 0 \end{bmatrix}, \quad (30)$$

where we have used Kronecker product properties to perform transpose. Finally, the gradients of loss function ℓ with respect to $\{\text{vec}(Q), q, \text{vec}(A), b, \text{vec}(G), h\}$ are computed by

$$\nabla_x \ell = \begin{bmatrix} \frac{\partial \ell}{\partial \text{vec}(Q)} \\ \vdots \\ \frac{\partial \ell}{\partial h} \end{bmatrix} = (\frac{\partial F(x, \theta^*)}{\partial x})^T \begin{bmatrix} d_z \\ d_\lambda \\ d_v \end{bmatrix}, \quad (31)$$

which can also be expressed compactly as

$$\begin{aligned} \frac{\partial \ell}{\partial \text{vec}(Q)} &= (z^* \otimes I_n) d_z & \frac{\partial \ell}{\partial q} &= I_n d_z = d_z \\ \frac{\partial \ell}{\partial \text{vec}(A)} &= (I_n \otimes v^*) d_z + (z^* \otimes I_m) d_v & \frac{\partial \ell}{\partial b} &= -I_m d_v = -d_v \\ \frac{\partial \ell}{\partial \text{vec}(G)} &= (I_n \otimes \lambda^*) d_z + (z^* \otimes D(\lambda^*)) d_\lambda & \frac{\partial \ell}{\partial h} &= -D(\lambda^*) d_\lambda. \end{aligned} \quad (32)$$

The second column of equation (32) already aligns with the second column of equation (8) in the paper. We now demonstrate that the first column of equation (32) can be further simplified to yield identical expressions too.

For $\frac{\partial \ell}{\partial \text{vec}(Q)}$, we expand $(z^* \otimes I_n)d_z$ as

$$\begin{bmatrix} z_1^* & 0 & \cdots & 0 \\ 0 & z_1^* & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & z_1^* \\ \vdots & \vdots & \vdots & \vdots \\ z_n^* & 0 & \cdots & 0 \\ 0 & z_n^* & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & z_n^* \end{bmatrix} \begin{bmatrix} (d_z)_1 \\ \vdots \\ (d_z)_n \end{bmatrix} = \begin{bmatrix} z_1^*(d_z)_1 \\ \vdots \\ z_1^*(d_z)_n \\ \vdots \\ z_n^*(d_z)_1 \\ \vdots \\ z_n^*(d_z)_n \end{bmatrix}. \quad (33)$$

Notice that this vector can be obtained by vectorizing a matrix of form

$$\begin{bmatrix} (d_z)_1 z_1^* & \cdots & (d_z)_1 z_n^* \\ \vdots & \ddots & \vdots \\ (d_z)_n z_1^* & \cdots & (d_z)_n z_n^* \end{bmatrix} = d_z(z^*)^T. \quad (34)$$

Therefore, $\frac{\partial \ell}{\partial \text{vec}(Q)} = \text{vec}(d_z(z^*)^T)$, and $\frac{\partial \ell}{\partial Q} = d_z(z^*)^T$ after reshaping. Moreover, the positive semidefiniteness of matrix Q implies symmetry, which further implies $\frac{\partial \ell}{\partial \text{vec}(Q)} = \frac{\partial \ell}{\partial \text{vec}(Q^T)}$. Applying almost the same derivation procedure as for $\frac{\partial \ell}{\partial \text{vec}(Q)}$, we obtain $\frac{\partial \ell}{\partial \text{vec}(Q^T)} = (I_n \otimes z^*)d_z$, which can be expanded as

$$\begin{bmatrix} z_1^* & \cdots & 0 \\ z_2^* & \cdots & 0 \\ \vdots & \ddots & \vdots \\ z_n^* & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & z_1^* \\ 0 & \cdots & z_2^* \\ \vdots & \ddots & \vdots \\ 0 & \cdots & z_n^* \end{bmatrix} \begin{bmatrix} (d_z)_1 \\ \vdots \\ (d_z)_n \end{bmatrix} = \begin{bmatrix} z_1^*(d_z)_1 \\ \vdots \\ z_n^*(d_z)_1 \\ \vdots \\ z_1^*(d_z)_n \\ \vdots \\ z_n^*(d_z)_n \end{bmatrix}, \quad (35)$$

Notice that this vector can be obtained by vectorizing a matrix of form

$$\begin{bmatrix} z_1^*(d_z)_1 & \cdots & z_1^*(d_z)_n \\ \vdots & \ddots & \vdots \\ z_n^*(d_z)_1 & \cdots & z_n^*(d_z)_n \end{bmatrix} = z^* d_z^T. \quad (36)$$

Therefore, $\frac{\partial \ell}{\partial \text{vec}(Q^T)} = \text{vec}(z^* d_z^T)$, and $\frac{\partial \ell}{\partial Q^T} = z^* d_z^T$ after reshaping. The symmetry suggests that $\text{vec}(d_z(z^*)^T) = \text{vec}(z^* d_z^T)$ and thus $d_z(z^*)^T = z^* d_z^T$. By decomposing square matrix $d_z(z^*)^T$ into its symmetric and skew-symmetric part, we obtain

$$\begin{aligned} d_z(z^*)^T &= \frac{1}{2}(d_z(z^*)^T + (d_z(z^*)^T)^T) + \frac{1}{2}(d_z(z^*)^T - (d_z(z^*)^T)^T) \\ &= \frac{1}{2}(d_z(z^*)^T + z^* d_z^T) + \frac{1}{2}(d_z(z^*)^T - z^* d_z^T) \\ &= \frac{1}{2}(d_z(z^*)^T + z^* d_z^T), \end{aligned} \quad (37)$$

from which we can conclude that $\frac{\partial \ell}{\partial Q} = \frac{1}{2}(d_z(z^*)^T + z^* d_z^T)$.

For $\frac{\partial \ell}{\partial \text{vec}(A)}$, by applying the same expansion and vectorization procedure as for $\frac{\partial \ell}{\partial \text{vec}(Q)}$, we obtain

$$\begin{aligned} \frac{\partial \ell}{\partial \text{vec}(A)} &= (I_n \otimes v^*)d_z + (z^* \otimes I_m)d_v \\ &= \text{vec}(v^* d_z^T) + \text{vec}(d_v(z^*)^T). \end{aligned} \quad (38)$$

Therefore, $\frac{\partial \ell}{\partial A} = v^* d_z^T + d_v(z^*)^T$ after reshaping.

For $\frac{\partial \ell}{\partial \text{vec}(G)}$, the derivation of the first terms follows the same procedure as before, while the expansion of the second term requires a slight modification. Expanding second term $(z^* \otimes D(\lambda^*))d_\lambda$ yields

$$\begin{bmatrix} z_1^* \lambda_1^* & 0 & \cdots & 0 \\ 0 & z_1^* \lambda_2^* & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & z_1^* \lambda_p^* \\ \vdots & \vdots & \vdots & \vdots \\ z_n^* \lambda_1^* & 0 & \cdots & 0 \\ 0 & z_n^* \lambda_2^* & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & z_n^* \lambda_p^* \end{bmatrix} \begin{bmatrix} (d_\lambda)_1 \\ \vdots \\ (d_\lambda)_p \end{bmatrix} = \begin{bmatrix} z_1^* \lambda_1^* (d_\lambda)_1 \\ \vdots \\ z_1^* \lambda_p^* (d_\lambda)_p \\ \vdots \\ z_n^* \lambda_1^* (d_\lambda)_1 \\ \vdots \\ z_n^* \lambda_p^* (d_\lambda)_p \end{bmatrix}, \quad (39)$$

which is vector that is obtained by vectorizing a matrix of form

$$\begin{aligned} \begin{bmatrix} z_1^* \lambda_1^* (d_\lambda)_1 & \cdots & z_n^* \lambda_1^* (d_\lambda)_1 \\ \vdots & \ddots & \vdots \\ z_1^* \lambda_p^* (d_\lambda)_p & \cdots & z_n^* \lambda_p^* (d_\lambda)_p \end{bmatrix} &= \begin{bmatrix} \lambda_1^* & 0 & \cdots & 0 \\ 0 & \lambda_2^* & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & \lambda_p^* \end{bmatrix} \begin{bmatrix} z_1^* (d_\lambda)_1 & \cdots & z_n^* (d_\lambda)_1 \\ \vdots & \ddots & \vdots \\ z_1^* (d_\lambda)_p & \cdots & z_n^* (d_\lambda)_p \end{bmatrix} \\ &= D(\lambda^*) d_\lambda (z^*)^T. \end{aligned} \quad (40)$$

Incorporating the first term, we obtain

$$\begin{aligned} \frac{\partial \ell}{\partial \text{vec}(G)} &= (I_n \otimes \lambda^*)d_z + (z^* \otimes D(\lambda^*))d_\lambda \\ &= \text{vec}(\lambda^* d_z^T) + \text{vec}(D(\lambda^*) d_\lambda (z^*)^T). \end{aligned} \quad (41)$$

Therefore, $\frac{\partial \ell}{\partial G} = \lambda^* d_z^T + D(\lambda^*) d_\lambda (z^*)^T$ after reshaping.

5 Gradient in Primal-Dual Interior Point Method

Section 3.1 of the paper closely follows the discussion on the primal-dual interior point method presented in lectures, and equation (9) and (10) of the paper can be derived by following the procedures outlined in the slides. Thus, we omit the derivation here and only focus on showcasing the derivation of equation (11).

We first obtain the symmetrized version of matrix K at optimality by scaling the 2_{nd} row by $D(1/s^*)$, which yields

$$K_{sym} = \begin{bmatrix} Q & 0 & G^T & A^T \\ 0 & D(\lambda^*/s^*) & I & 0 \\ G & I & 0 & 0 \\ A & 0 & 0 & 0 \end{bmatrix}. \quad (42)$$

Here, we have used the fact that $D(\lambda^*)D(1/s^*) = D(\lambda^*/s^*)$ and $D(s^*)D(1/s^*) = I$. The introduction of extra slack variable s does not prevent us from using equation (8) of the paper to compute backpropagated gradients. This can be seen by noticing

$$K_{sym} \begin{bmatrix} d_z \\ d_s \\ \tilde{d}_\lambda \\ d_v \end{bmatrix} = \begin{bmatrix} Qd_z + G^T \tilde{d}_\lambda + A^T d_v \\ D(\lambda^*/s^*)d_s + \tilde{d}_\lambda \\ Gd_z + d_s \\ Ad_z \end{bmatrix}. \quad (43)$$

Substituting $\tilde{d}_\lambda$ with $D(\lambda^*)d_\lambda$, we obtain

$$\begin{bmatrix} Qd_z + G^T D(\lambda^*)d_\lambda + A^T d_v \\ d_s + (D(\lambda^*/s^*))^{-1} D(\lambda^*)d_\lambda \\ Gd_z + d_s \\ Ad_z \end{bmatrix} = \begin{bmatrix} -\frac{\partial \ell}{\partial z_{i+1}} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (44)$$

from which we can see that the 2_{nd} and 3_{rd} rows can be represented together by

$$\begin{aligned} Gd_z + d_s &= Gd_z - (D(\lambda^*/s^*))^{-1} D(\lambda^*)d_\lambda \\ &= Gd_z - D(s^*)d_\lambda \\ &= Gd_z + D(-s^*)d_\lambda \\ &= Gd_z + D(Gz^* - h)d_\lambda, \end{aligned} \quad (45)$$

and $z^* = z_{i+1}$. This means that solving equation (11) of the paper is equivalent of solving

$$\begin{bmatrix} Qd_z + G^T D(\lambda^*)d_\lambda + A^T d_v \\ Gd_z + D(Gz^* - h)d_\lambda \\ Ad_z \end{bmatrix} = \begin{bmatrix} -\frac{\partial \ell}{\partial z^*} \\ 0 \\ 0 \end{bmatrix}, \quad (46)$$

which is basically the equation (7) of the paper with terms rearranged.

The use of primal-dual interior point method and symmetrized version of matrix K allows a batch of quadratic optimization problems with same form of problem (1) to be solved in parallel. The gradients computed during this optimization process can be reused to compute backward gradients using equation (11) in the paper. These features make OptNet training faster and more resource-efficient.

The remaining part of Section 3 is relatively more detailed than the other parts. The three theorems are proved with great details in the Supplementary Material section, we therefore refer readers to the paper for more details.

References

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